Dict

In [3]:
```
d = {}
type(d)
```

Out[3]:
```
dict
```

In [5]:
```
d1 = {'a' : 1, 'b' : 2, 3 : True, 4.3 : 1+2j}
d1
```

Out[5]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [7]:
```
d1.keys()
```

Out[7]:
```
dict_keys(['a', 'b', 3, 4.3])
```

In [9]:
```
d1.values()
```

Out[9]:
```
dict_values([1, 2, True, (1+2j)])
```

In [11]:
```
d1.items()
```

Out[11]:
```
dict_items([('a', 1), ('b', 2), (3, True), (4.3, (1+2j))])
```

In [15]:
```
d2 =d1.copy()
d2
```

Out[15]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [17]:
```
print(d1)
print(d2)
```

```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [19]:
```
d1[4.3]
```

Out[19]:
```
(1+2j)
```

In [21]:
```
d2[3]
```

Out[21]:
```
True
```

In [23]:
```
mydict3 = {1 :'one', 2:'two', 'A':{'Name': 'shraddha', 'Age':22},'B' : ('Bat','Cat'
mydict3
```

Out[23]:
```
{1: 'one',
 2: 'two',
 'A': {'Name': 'shraddha', 'Age': 22},
 'B': ('Bat', 'Cat', 'Hat')}
```

In [25]:
```
mydict3.items()
```

Out[25]:
```
dict_items([(1, 'one'), (2, 'two'), ('A', {'Name': 'shraddha', 'Age': 22}), ('B',
('Bat', 'Cat', 'Hat'))])
```

In [27]:
```
keys = {'a','b','c','d'}

mydict3 = dict.fromkeys(keys)
mydict3
```

Out[27]:
```
{'b': None, 'a': None, 'c': None, 'd': None}
```

In [29]:
```
keys = {'a','b','c','d'}
values = 10
mydict3 = dict.fromkeys(keys,values)
mydict3
```

Out[29]:
```
{'b': 10, 'a': 10, 'c': 10, 'd': 10}
```

In [31]:
```
keys = {'a','b','c','d'}
values =[10,20,30]
mydict3 = dict.fromkeys(keys,values)
mydict3
```

Out[31]:
```
{'b': [10, 20, 30], 'a': [10, 20, 30], 'c': [10, 20, 30], 'd': [10, 20, 30]}
```

In [33]:
```
values.append(40)
mydict3
```

Out[33]:
```
{'b': [10, 20, 30, 40],
 'a': [10, 20, 30, 40],
 'c': [10, 20, 30, 40],
 'd': [10, 20, 30, 40]}
```

In [35]:
```
mydict3.get('d')
```

Out[35]:
```
[10, 20, 30, 40]
```

In [37]:
```
d1
```

Out[37]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [39]:
```
d1[10]= 'ten'
d1
```

Out[39]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j), 10: 'ten'}
```

In [41]:
```
d1.pop('b')
```

Out[41]:
```
2
```

In [43]:
```
d1.popitem()
```

Out[43]: (10, 'ten')

In [45]:
```
all(d2)
```

Out[45]: True

In [47]:
```
any(d2)
```

Out[47]: True

Range

In [50]:
```
range(10)
```

Out[50]: range(0, 10)

In [54]:
```
list(range(1,10))
```

Out[54]: [1, 2, 3, 4, 5, 6, 7, 8, 9]

In [56]:
```
list(range(1,10,3))
```

Out[56]: [1, 4, 7]

In [58]:
```
r = range(10)
r
```

Out[58]: range(0, 10)

In [60]:
```
for i in r:
    print(i)
```

```
0
1
2
3
4
5
6
7
8
9
```